

GOURI ANIL

Baton Rouge, LA

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[Website](#)

EDUCATION

LOUISIANA STATE UNIVERSITY

PhD Candidate

Department of Oceanography and Coastal Sciences

CGPA: 4.18

Baton Rouge, LA

Aug 2022 – Aug 2026

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

MS in Physical Oceanography

CGPA: 9.69 (on 10)

Kerala, India

Aug 2019 - Jun 2021

MAR ATHANASIOS COLLEGE OF ARTS AND SCIENCES

BS in Physics

CGPA: 9.43 (on 10)

Kerala, India

Jun 2016 - May 2019

RESEARCH EXPERIENCE

LOUISIANA STATE UNIVERSITY

PhD Candidate

Baton Rouge, LA

Aug 2022 - Present

Stratospheric Aerosol Injection (SAI) and Marine Ecosystems: Impacts with Emphasis on Coral Reefs

Advisor: Dr. Cheryl S Harrison

Committee: Drs. Eurico D'Sa, Kristine DeLong, Daniel Holstein, Junhong Liang, Gaurav Kandlikar

Project 1: Impacts of Timing and Intensity of SAI on Coral Reefs

Project 2: Global Marine Climate Exposure under SAI

Project 3: Building an Observation-Based Model to Predict Coral Bleaching from Both Heat and Light Stress to Assess Impacts of SAI

INDIAN NATIONAL CENTRE FOR OCEAN INFORMATION SYSTEMS (INCOIS)

Graduate Researcher

Advisor: Dr. Abhisek Chatterjee (Senior Scientist, INCOIS)

Hyderabad, India

Dec 2020 - May 2021

Understanding Marine Heatwaves in the Arabian Sea

INDIAN INSTITUTE OF TECHNOLOGY

Undergraduate Summer Research Fellow

Advisor: Dr. Varun Bhalerao

Mumbai, India

May 2018 - Jul 2018

X-Ray data analysis of a pulsar Centaurus X-3 using NASA HEASoft

EXTRAMURAL FUNDING

2025 **Environmental Defense Fund**, Concept Lead, Lead Proposal Writer and Key Personnel (Funded)

Preventing Coral Bleaching Through SAI: An Observation-Based Model to Predict Coral Bleaching from Both Heat and Light Stress to Assess Impacts of Stratospheric Aerosol Injection

PI: Dr. Cheryl S. Harrison, Collaborator: Dr. Thomas Oliver

PUBLICATIONS

[Google Scholar](#)

Published

A. Chatterjee, **G. Anil**, L. R. Shenoy, *Marine Heatwaves in the Arabian Sea*, Ocean Science, 2022,

DOI: 10.5194/os-18-639-2022

K. E., Roberts, T. Rohr, ..., **G. Anil**, ...C.S. Harrison (2026). *Potential Impacts of Climate Interventions on Marine Ecosystems*. Reviews of Geophysics, 64(1), e2024RG000876.

<https://doi.org/10.1029/2024RG000876>

Accepted

L. Kounta, L. Luo, **G. Anil**, D. Hueholt, C. S. Harrison, D. Visoni, M. Tye, T. Felgenhauer, A. T. Gaye, P.L. Zarnetske, *Climate Intervention through Stratospheric Aerosol Injection may partially mitigate marine heatwaves*, ESS Open Archive. November 29, 2025. DOI: 10.22541/essoar.176442822.27571009/v1

In Review

G. Anil, C.S. Harrison, J.A. Kleypas, D. Holstein, T.A. Oliver, V. Garza, K.E. Roberts, T. Felgenhauer, D. Jayewardene, *Impacts of Timing and Level of Stratospheric Aerosol Injection on Coral Thermal Bleaching*, ESS Open Archive. December 17, 2025. DOI: 10.22541/essoar.176599087.79155141/v1

In Prep

G. Anil, K. E., Roberts, D. Boyce, V. Garza, C.S. Harrison, *Could Stratospheric Aerosol Injection Limit Climate Exposure for Global Marine Ecosystems?*, Nature Climate Change

H.C. Olivarez, C. S. Harrison, J. S. Coupe, B. Kravitz, **G. Anil**, V. Garza, N. S. Lovenduski, *Impacts of Climate Intervention on Ocean Acidification and the Aragonite Saturation Horizon*, Environmental Research Letters.

E.K. Asante, L. Gouvea, **G. Anil**, J. Assis, K. Cavanaugh, C.S. Harrison, *Projected Impacts of Stratospheric Aerosol Injection on Global Mangrove Distribution, Aboveground Biomass, and Carbon Stocks*

PRESENTATIONS

ORAL

2026	International Coral Reef Symposium 2026, New Zealand (July)
2026	16 th Geoengineering Model Intercomparison Project (GeoMIP) Annual Meeting, Japan
2025	AGU Fall Meeting, USA
	LSU 3-Minute Thesis Competition Finals, USA
	LSU College of Coast and Environment Annual Graduate Student Symposium, USA
2024	Reef Futures Symposium, Mexico

POSTERS

- 2026 16th Annual GeoMIP Annual Meeting
Coupled Model Intercomparison Project Community Workshop (Presenter: C S. Harrison)
- 2025 AGU Fall Meeting
- 2024 LSU Graduate Research Conference
- 2024 AGU Ocean Sciences Meeting

MEDIA COVERAGE ON GRADUATE RESEARCH

- 2025 Could Stratospheric Aerosol Injection Help Save Corals from Bleaching? *Eos*, [Article](#)
- 2023 Marine heat waves may be favouring more cyclones in the Arabian Sea: INCOIS Scientists, *The Hindu*, [Article](#)
- 2022 'At least 20 marine heatwave days in the Arabian Sea during last decade', *The Times of India*, [Article](#)
- 2022 The Arabian Sea faces an increased frequency and duration of marine heatwave days, finds study, *Mongabay-India*, [Article](#)

AWARDS AND FELLOWSHIPS

- 2025 Travel Award to attend the 16th Geoengineering Model Intercomparison Project Meeting, Tokyo
- 2025 NCAR High-Performance Computing Large Allocation for Climate Intervention Simulations
- 2025 Travel Award to attend the 2025 Degrees Global Forum, South Africa
- 2024 Graduate Student Organization Travel Award
- 2023 Travel Award to attend the Scripps Marine Geoengineering Workshop
- 2022 Graduate Excellence Award for Academic Achievement
- 2019 Indian Academy of Sciences Summer Research Fellowship
- 2018 Indian Academy of Sciences Summer Research Fellowship

PEER REVIEW

- 2025 Co-reviewer for Journal of Geophysical Research – Oceans
- 2025 Reviewer for American Academy of Arts & Sciences Solar Geoengineering Primer

TEACHING AND MENTORING

- 2026 Teaching Assistant and Guest Lecturer, Gulf of Mexico (Spring Semester)
- 2026 Undergraduate Honors Thesis Mentor (Spring Semester)
- 2024 Teaching Assistant, Introduction to Oceanography (Fall Semester)
- 2023 Teaching Assistant, Introduction to Oceanography (Fall Semester)

WORKSHOPS

- 2024 Community Climate Intervention Strategies (CCIS) Workshop, Notetaker
- 2023 Scripps Marine Geoengineering Scenario Design Workshop, Rapporteur
- 2019 International Conference on Frontiers in Marine Science, Volunteer and Session Host

OUTREACH

- 2024 Student Host for Ocean Commotion Outreach Event by LSU
- 2024 Student Host for LSU College of Coast and Environment Open House
- 2024 Coast and Environment Graduate Organization Secretary
- 2018 Volunteer and Team Lead, Guardians of Dreams Non-profit, India
- 2016 Volunteer and HR Team Member, The Art Outreach Society Non-profit, India

TECHNICAL SKILLS

Programming: Python, R, Machine Learning and Bayesian Statistical Tools

Modeling: Experience running Community Earth System Model Simulations, Model Data Processing

Climate Data Processing and Visualization: netCDF, xarray, geopandas, matplotlib, CDO

Spatial Tools: QGIS